

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE CODE: ITA 06

COURSE NAME: MACHINE LEARNING

COURSE OUTCOME COVERED

CO2: *Neural Network Representation, Perceptron Model, Multilayer Perceptrons, Backpropagation Algorithm, Deep Neural Networks, Genetic Algorithms, Hypothesis Space Search, Genetic Programming, Models of Evaluation and Learning (BL1, BL2)*

Assessment Tool Weightage: 40%

QUESTIONS: 15

Total Marks:25

Annexure A

Analytical Problem Solving

Code of Conduct:

I Prudhvi Raj.B (192324274) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: Prudhvi raj.b

Annexure A

Short Answer Test

1, A simple perceptron model is used to classify whether a student passes or fails based on two input features, namely study hours and attendance. The weights associated with these inputs are 0.6 and 0.4 respectively, and the bias value is -0.5 . For a given student with study hours equal to 2 and attendance equal to 1, compute the net input to the perceptron and apply the step activation function to determine the final output. Based on the result, conclude whether the student passes or fails. (5 Marks)

2. In a neural network, a single neuron is trained using the backpropagation learning rule. The initial weight is 0.5 and the learning rate is 0.1. For an input value of 2, the actual output produced by the neuron is 0.6, whereas the target output is 1. Calculate the error in the output and determine the weight update using the delta rule. Hence, compute the new updated weight after one iteration of training. (5 Marks)

3. A deep learning model is developed for disease prediction and its performance is evaluated using a confusion matrix. The model yields 180 true positives, 150 true negatives, 20 false positives, and 50 false negatives. Using these values, compute the accuracy, precision, sensitivity (recall), and specificity of the model. Based on the calculated results, discuss whether the model is reliable for medical diagnosis, considering the importance of minimizing diagnostic errors. (5 Marks)

4. A Genetic Algorithm is applied to maximize the function $f(x) = x^2$ using an initial population consisting of the values 2, 4, 6, and 8. Determine the fitness value for each individual in the population and select the best candidates based on their fitness scores. Perform a crossover operation by exchanging the least significant bits of the selected individuals to generate new offspring. Additionally, explain the role of mutation in improving the diversity and effectiveness of the solution in Genetic Algorithms. (5 Marks)

5. During the training of a deep neural network model, the training and validation accuracies are observed over multiple epochs. The training accuracy increases steadily from 60% to 95%, while the validation accuracy initially increases from 58% to 72% but then decreases to 68% in later epochs. Analyse this trend and identify the issue affecting the model after a certain point in training. Explain why this issue occurs in neural networks and suggest two effective techniques to address it. Also, discuss the significance of validation accuracy in evaluating the performance of a learning model. (5 Marks)

6. A perceptron model is used to classify whether a loan is approved based on two features: income and credit score. The weights are 0.7 and 0.5, and the bias is -0.6 .

For an applicant with income = 1.5 and credit score = 2, compute the net input and apply the step activation function. Determine whether the loan is approved or rejected. (5 Marks)

7. In a neural network, a single neuron uses the delta learning rule. The initial weight is 0.4, learning rate is 0.2, and input is 3. The actual output is 0.5, while the target output is 0.9. Calculate the error, determine the weight update, and compute the new weight after one iteration. (5 Marks)

8. A classification model produces the following confusion matrix values: True Positives = 120, True Negatives = 200, False Positives = 30, and False Negatives = 50. Compute accuracy, precision, recall (sensitivity), specificity, and F1-score. Based on these results, evaluate whether the model is suitable for applications where false negatives are critical. (5 Marks)

9. A Genetic Algorithm is used to maximize the function $f(x) = 2x + 3$ for an initial population of values {1, 3, 5, 7}. Compute the fitness of each individual and select the top two candidates. Perform a single-point crossover to generate offspring and explain how mutation can help avoid premature convergence. (5 Marks)

10. During training of a neural network, the training loss decreases continuously, while validation loss decreases initially and then starts increasing after several epochs. Identify the issue occurring in the model. Explain the reason behind this behavior and suggest two techniques such as regularization or early stopping to improve generalization performance. (5 Marks)

11. A perceptron model is used to classify whether a customer will purchase a product based on two features: advertisement exposure and income level. The weights are 0.5 and 0.3, and the bias is -0.4. For a customer with advertisement exposure = 2 and income level = 1, compute the net input to the perceptron and apply the step activation function. Determine whether the customer makes a purchase or not. (5 Marks)

12. In a neural network, a neuron is trained using the delta learning rule. The initial weight is 0.6, the learning rate is 0.15, and the input is 2.5. The actual output is 0.7, while the target output is 1. Calculate the error, determine the weight update, and compute the new updated weight after one iteration. (5 Marks)

13. A classification model produces the following confusion matrix values: True Positives = 140, True Negatives = 180, False Positives = 25, and False Negatives = 35. Compute accuracy, precision, recall (sensitivity), specificity, and F1-score. Based on the results, analyze whether the model is suitable for applications such as fraud detection where false positives are costly. (5 Marks)

14. A Genetic Algorithm is applied to maximize the function $f(x) = 3x + 2$ for an initial population {2, 5, 7, 9}. Compute the fitness value for each individual and select the best candidates. Perform a crossover operation to generate new offspring and explain how mutation helps in maintaining diversity and avoiding local optima.

(5 Marks)

15. During the training of a deep learning model, the training accuracy increases from 65% to 97%, while validation accuracy increases from 63% to 75% and then fluctuates around 72%. Identify the issue present in the model. Explain why this happens and suggest techniques such as dropout and data augmentation to improve the model's performance. Also, discuss the importance of validation metrics in model evaluation. (5 Marks)

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary (5 Marks)	Level 3 – Proficient (4 Marks)	Level 2 – Developing (2–3 Marks)	Level 1 – Beginning (0–1 Mark)
Conceptual Understanding	25	Demonstrates clear and complete understanding of the concept	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor or incorrect understanding
Accuracy of Answer	25	Completely correct answer with no errors	Mostly correct with minor errors	Partially correct with noticeable errors	Incorrect or irrelevant answer
Application of Knowledge	20	Correctly applies concepts to solve the question/problem	Applies concepts with minor mistakes	Limited or partially correct application	No or incorrect application
Completeness of Response	15	Covers all required parts of the question	Covers most parts with minor omissions	Covers only some parts	Incomplete or missing response

Clarity & Presentation	15	Clear, concise, and well-organized answer	Understandable with minor issues	Somewhat unclear or poorly structured	Difficult to understand
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Question 1

Perceptron for pass/fail classification (weights 0.6, 0.4; bias -0.5; study hours = 2, attendance = 1).

Given: $w_1 = 0.6$, $w_2 = 0.4$, bias = -0.5, x_1 (study hours) = 2, x_2 (attendance) = 1

Step 1: Compute net input

$$\begin{aligned}\text{Net input} &= (w_1 \times x_1) + (w_2 \times x_2) + \text{bias} \\ &= (0.6 \times 2) + (0.4 \times 1) + (-0.5) \\ &= 1.2 + 0.4 - 0.5 = \mathbf{1.1}\end{aligned}$$

Step 2: Apply step activation function

$$f(\text{net}) = 1, \text{ if } \text{net} \geq 0; \quad f(\text{net}) = 0, \text{ if } \text{net} < 0$$

Since net input = 1.1 ≥ 0 , output = 1

Conclusion: Since the perceptron output is 1, the student is classified as **PASS**.

Question 2

Backpropagation / delta rule: initial weight = 0.5, learning rate = 0.1, input = 2, actual output = 0.6, target output = 1.

Step 1: Compute the error

$$\text{Error (E)} = \text{Target} - \text{Actual Output} = 1 - 0.6 = \mathbf{0.4}$$

Step 2: Compute weight update using the Delta Rule

$$\Delta w = \eta \times E \times x$$

$$\Delta w = 0.1 \times 0.4 \times 2 = \mathbf{0.08}$$

Step 3: Compute the updated weight $w(\text{new})$

$$= w(\text{old}) + \Delta w = 0.5 + 0.08 = \mathbf{0.58}$$

Conclusion: After one iteration of training, the updated weight is **0.58**.

Question 3

Confusion matrix for disease prediction: TP = 180, TN = 150, FP = 20, FN = 50.

Total samples = TP + TN + FP + FN = 180 + 150 + 20 + 50 = 400

Metric	Formula	Calculation	Result
Accuracy	$(\text{TP} + \text{TN}) / \text{Total}$	$(180 + 150) / 400$	82.5%
Precision	$\text{TP} / (\text{TP} + \text{FP})$	$180 / 200$	90%
Sensitivity (Recall)	$\text{TP} / (\text{TP} + \text{FN})$	$180 / 230$	78.26%
Specificity	$\text{TN} / (\text{TN} + \text{FP})$	$150 / 170$	88.24%

Discussion on reliability: The model shows high accuracy (82.5%) and strong precision (90%), meaning most positive predictions are correct. However, its sensitivity/recall (78.26%) is comparatively lower, meaning 50 diseased patients were missed (false negatives). In medical diagnosis, minimizing false negatives is critical since a missed diagnosis can delay treatment and endanger the patient. Therefore, although the model performs reasonably well overall, it is **not fully reliable** for critical medical diagnosis unless recall/sensitivity is further improved (e.g., by adjusting the decision threshold or using cost-sensitive learning).

Question 4

Genetic Algorithm to maximize $f(x) = x^2$; initial population = {2, 4, 6, 8}.

Step 1: Compute fitness ($f(x) = x^2$) for each individual

x	2	4	6	8
Fitness = x^2	4	16	36	64

Step 2: Select best candidates

The individuals with the highest fitness are **6 (fitness = 36)** and **8 (fitness = 64)**; these are selected as parents for reproduction.

Step 3: Crossover (exchange least significant bits)

Binary representation (4-bit): 6 = 0110, 8 = 1000

Exchanging the last 2 (least significant) bits of the two parents:

Parent 6 (0110) → takes last 2 bits of 8 (00) → Offspring 1 = 0100 = **4**

Parent 8 (1000) → takes last 2 bits of 6 (10) → Offspring 2 = 1010 = **10**

New offspring generated: 4 and 10

Role of mutation: Mutation randomly flips a bit in an offspring's chromosome with a small probability. It introduces new genetic material that crossover alone cannot produce, maintains diversity within the population, prevents the algorithm from converging prematurely to a local optimum, and allows exploration of unvisited regions of the search space — improving the overall effectiveness of the GA in finding the global optimum.

Question 8

Confusion matrix: TP = 120, TN = 200, FP = 30, FN = 50.

Total samples = 120 + 200 + 30 + 50 = 400

Metric	Formula	Calculation	Result
Accuracy	$(TP+TN)/\text{Total}$	$(120+200)/400$	80%
Precision	$TP/(TP+FP)$	$120/150$	80%
Recall (Sensitivity)	$TP/(TP+FN)$	$120/170$	70.59%
Specificity	$TN/(TN+FP)$	$200/230$	86.96%
F1-Score	$2 \times (P \times R) / (P + R)$	$2 \times (0.8 \times 0.7059) / (1.5059)$	75.01%

Evaluation: With a recall of only 70.59%, the model misses close to 30% of actual positive cases (50 false negatives). For applications where false negatives are critical (e.g., disease detection, fraud/safety alerts), this recall is **not sufficient**. The model would need to be tuned (e.g., lowering the classification threshold or using cost-sensitive training) to reduce false negatives before being considered suitable for such critical applications.